Business Requirements

Overview

The **OptiCrop: Smart Agricultural Production Optimization Engine** should provide an intelligent and user-friendly platform that helps farmers make better crop selection decisions based on soil and environmental conditions.

Functional Requirements

- Analyze agricultural data such as Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, rainfall, soil pH, and seasonal conditions.
- Recommend the most suitable crop based on the input parameters.
- Provide a simple, responsive, and easy-to-use web interface.
- Display crop prediction results quickly and accurately.
- Integrate the machine learning model with the web application for real-time predictions.

Technical Requirements

- Support machine learning models such as K-Nearest Neighbors (KNN), Logistic Regression, Decision Tree, Random Forest, and K-Means Clustering.
- Perform data preprocessing and feature scaling before prediction.
- Evaluate model performance to ensure accurate recommendations.
- Present prediction results in a clear and understandable format.

Non-Functional Requirements

- Ensure fast and reliable system performance.
- Maintain scalability for future enhancements.
- Support easy integration with larger datasets and smart farming technologies.
- Help optimize the use of water, fertilizers, and soil nutrients while reducing crop failure risks.